

On Algebraic Invariances Appearing in Continued Fractions and other Polynomial Recursive Objects

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Recursive Objects & Polynomials

A ‘polynomial recursive object’ shall be any infinite, *periodic* recursion of algebraic operations that can be unraveled by the solution of a polynomial equation. One such example is the following classic continued fraction:

$$X = 1 + \frac{1}{1 + \frac{1}{1 + \dots}} \Rightarrow X = 1 + \frac{1}{X} \Rightarrow X^2 - X - 1 = 0$$

Where by noticing how X appears ‘within itself’, an otherwise infinite, cascading object can be swiftly associated with a short, compact polynomial.

To find the numerical value of X one would of course solve its associated polynomial, with this being a consequence of the Fixed Point Iteration Theorem for the contractive map $x \mapsto 1 + 1/x, x \geq 1$.

Here, we shall treat recursive objects as *representations* of their associated polynomial who’s roots, candidate values for the numerical result, will not be specified explicitly.

For example, to each linear polynomial $ax + b$ we can ascribe the following recursive representation;

$$X = b + (1 + a)(b + (1 + a)(b + (1 + a)(\dots))) \Rightarrow X = b + (1 + a)X \Rightarrow aX + b = 0$$

To each quadratic polynomial $x^2 = ax + b$ we could associate it with infinitely nested square roots;

$$X = \sqrt{b + a\sqrt{b + a\sqrt{b + a\sqrt{\dots}}}} \Rightarrow X^2 = b + aX$$

Or, as above, it can be represented by a periodic continued fraction;

$$X = a + \frac{b}{a + \frac{b}{a + \dots}} \Rightarrow X = a + \frac{b}{X} \Rightarrow X^2 = aX + b$$

Continued fractions are *especially* ‘quadratic’ in the sense that their associated polynomial is of degree (at most) two no matter the ‘*depth*’ of their recursion;

$$X = a_1 + \frac{b_1}{\ddots + \frac{\ddots}{a_{n-1} + \frac{b_{n-1}}{a_n + \frac{b_n}{X}}}} \Rightarrow X = a_1 + \frac{b_1}{\ddots + \frac{\ddots}{a_{n-1} + \frac{b_{n-1}X}{a_n X + b_n}}} \Rightarrow \dots \Rightarrow X = \frac{\alpha_1 X + \alpha_2}{\beta_1 X + \beta_2}$$

Where the rational function keeps ‘flip-flopping’ and always retaining degree of at most 1 in the numerator and denominator. Yet another recursive representation for quadratics is the ‘doubly continued fraction’;

$$X^2 = aX + b \Leftrightarrow X = \frac{b + X}{1 - a + X} = \frac{b + \frac{b+X}{1-a+X}}{1 - a + \frac{b+X}{1-a+X}} \text{ etc.}$$

While it is true that the quickest route to determining a recursive object’s numerical value is to solve its associated polynomial, we shall soon see how *properties* and *symmetries* of this polynomial carry even greater elegance.

Root & Coefficient Mappings

As the *entries* in a recursive object correspond to the *coefficients* of its associated polynomial, and as the *numerical value* of the object corresponds to the *roots* of this polynomial, the mappings of polynomial *coefficients* under operations of the *roots* directly characterizes the effects of these operations on the recursive object.

Perhaps the simplest operation on the roots of a polynomial is to scale each by a factor of k . This map, $\mathcal{M}_{\times k}$, would take the quadratic $x^2 + Ax + B$ and return $x^2 + Akx + Bk^2$ and hence we may define;

$$\mathcal{M}_{\times k} : [1, A, B] \rightarrow [1, Ak, Bk^2]$$

Since these new coefficients can be *re-associated* with their respective recursive object, the action of $\mathcal{M}_{\times k}$ can be visualized as an action on, for example, infinite fractions:

$$X = a + \frac{b}{a + \frac{b}{a + \dots}} \iff kX = ak + \frac{bk^2}{ak + \frac{bk^2}{ak + \dots}}$$

Without a hint of induction in sight.

As a less trivial example, the mapping of the quadratic $x^2 + Ax + B$ corresponding to shifting each of the roots by $+t$ would be $(x-t)^2 + A(x-t) + B$, or;

$$\mathcal{M}_{+t} : [1, A, B] \rightarrow [1, A - 2t, B - At + t^2]$$

As before, we can interpret this mapping on the coefficients as a way to apply transformations to a recursive object without disturbing its structure;

$$\sqrt{b + a\sqrt{b + a\sqrt{b + a\sqrt{\dots}}}} + t = \sqrt{(b - at - t^2) + (a + 2t)\sqrt{(b - at - t^2) + (a + 2t)\sqrt{(b - at - t^2) + (a + 2t)\sqrt{\dots}}}}$$

Yet another transformation on the roots of a quadratic is the *squaring* operation given by;

$$\mathcal{M}_{\square^2} : [1, A, B] \rightarrow [1, 2B - A^2, B^2]$$

Which can again be visualized through additional identities given by the continued fraction representation;

$$\left(a + \frac{b}{a + \frac{b}{a + \dots}}\right)^2 = a^2 + 2b + \frac{-b^2}{a^2 + 2b + \frac{-b^2}{a^2 + 2b + \dots}}$$

It should be noted that each of the maps have an intuitive inverse in terms of *undoing* the respective root arithmetic; i.e. $\mathcal{M}_{+t}^{-1} = \mathcal{M}_{-t}$, $\mathcal{M}_{\times k}^{-1} = \mathcal{M}_{\div k}$, etc.

And in this spirit, the action of taking the squares and square-roots of the roots of a quadratic $x^2 - Ax - B$ can be similarly captured in the coefficients;

$$\mathcal{M}_{\sqrt{}} = \mathcal{M}_{\square^2}^{-1} : [1, A, B] \rightarrow [1, \pm\sqrt{\pm 2\sqrt{B} - A}, \pm\sqrt{B}]$$

For appropriate choice of branch. When applied to the continued fraction representation, this gives the peculiar;

$$\sqrt{a + \frac{b}{a + \frac{b}{a + \dots}}} = \sqrt{a + 2\sqrt{-b}} + \frac{-\sqrt{-b}}{\sqrt{a + 2\sqrt{-b}} + \frac{-\sqrt{-b}}{\sqrt{a + 2\sqrt{-b}} + \dots}}$$

But undoubtedly the most delightful of the polynomial root mappings is the *reciprocal* action, which although seemingly non-linear requires only *reversing* the order of the coefficients;

$$\mathcal{M}_{1/} : [1, A_1, A_2, \dots, A_n] \rightarrow [A_n, \dots, A_2, A_1, 1] = \left[1, \dots, \frac{A_2}{A_n}, \frac{A_1}{A_n}, \frac{1}{A_n}\right]$$

For a change of pace, let's apply this to a general cubic representation;

$$X = -\frac{a}{2} + \sqrt{\frac{a^2}{4} - b - \frac{c}{-\frac{a}{2} + \sqrt{\frac{a^2}{4} - b - \frac{c}{-\frac{a}{2} + \sqrt{\frac{a^2}{4} - b - \dots}}}}} \Rightarrow X^3 + aX^2 + bX + c$$

$$\frac{1}{X} = -\frac{bc^{-1}}{2} + \sqrt{\frac{b^2c^{-2}}{4} - ac^{-1} - \frac{c^{-1}}{-\frac{bc^{-1}}{2} + \sqrt{\frac{b^2c^{-2}}{4} - ac^{-1} - \frac{c^{-1}}{-\frac{bc^{-1}}{2} + \sqrt{\frac{b^2c^{-2}}{4} - ac^{-1} - \dots}}}}} \Rightarrow X^3 + aX^2 + bX + c$$

Unsurprisingly, when expressed in this infinite, clunky, nested radical form, the underlying *simplicity* of reversing coefficients in a polynomial is much harder to detect.

It would of course be impossible to enumerate all such curious identities which could be formed by combining an arbitrary operation on polynomial roots with any possible recursive-object-representation.

Instead, I mean to illustrate how the root $\Leftrightarrow$ recursive-object connection can be visualized and intuited to motivate many more such representations.

Generators & Invariance

With an understanding of how algebraic operations on the *roots* of associated polynomials translate to equivalent operations on the *recursive objects* they represent, we now explore a deeper *intuition* behind why the numerical value for an object can be taken from the corresponding set of roots.

Suppose the recursive object X is formed from the periodic composition of algebraic operations $f_1, \dots, f_n$ such that $X = f_1(f_2(\dots f_n(f_1(f_2(\dots)))) \dots)$. We have *defined* the associated polynomial to be the ‘unraveled’ version of $X = f_1(f_2(\dots f_n(X) \dots))$ and understand its roots to be candidate values for X by the fixed-point iteration.

Our key observation, which should not be at all surprising, is that the roots of the polynomial associated to X are invariant under the same *finite* coefficient mapping $\mathcal{M}_{f_1} \circ \mathcal{M}_{f_2} \circ \dots \mathcal{M}_{f_n}$ as X .

In the opening example of $X = 1 + \frac{1}{1 + \frac{1}{1 + \dots}}$, we have that $f_1(x) = 1 + x$, $f_2(x) = 1/x$, and $x = 1 + 1/x \Rightarrow x^2 - x - 1 = 0 \Leftrightarrow [1, -1, -1]$ so we note that;

$$\mathcal{M}_{+1} \circ \mathcal{M}_{1/} \circ [1, -1, -1] = \mathcal{M}_{+1} \circ [-1, -1, 1] = \mathcal{M}_{+1} \circ [1, 1, -1] = \mathcal{M}_{+1} \circ [1, 1-2, -1-1+1^2] = [1, -1, -1]$$

This is to say that while the *value* of X is a fixed point of $f_1 \circ f_2$, the *coefficients* of X 's associated polynomial are fixed points of $\mathcal{M}_{f_1} \circ \mathcal{M}_{f_2}$.

Since $X = \sqrt{1 + \sqrt{1 + \sqrt{\dots}}}$ also has a coefficient representation of $[1, -1, -1]$ this must be invariant under the mapping $f_1(x) = \sqrt{x}$, $f_2(x) = 1 + x$ too;

$$\mathcal{M}_{\sqrt{}} \circ \mathcal{M}_{+1} \circ [1, -1, -1] = \mathcal{M}_{\sqrt{}} \circ [1, -1-2, -1-(-1)+1^2] = \mathcal{M}_{\sqrt{}} \circ [1, -3, 1] = [1, -\sqrt{-2\sqrt{1}+3}, -\sqrt{1}] = [1, -1, -1]$$

Where the negative branch has been taken in each case. Walking through these mappings for a general continued fraction, say, $X = t + \frac{k}{t + \frac{k}{t + \dots}}$ $\Rightarrow X = t + \frac{k}{X} \Rightarrow X^2 - tX - k = 0$ gives the same notion coefficient invariance;

$$\begin{aligned} \mathcal{M}_{+t} \circ \mathcal{M}_{\times k} \circ \mathcal{M}_{1/} \circ [1, -t, -k] &= \mathcal{M}_{+t} \circ \mathcal{M}_{\times k} \circ [-k, -t, 1] = \mathcal{M}_{+t} \circ \mathcal{M}_{\times k} \circ \left[1, \frac{t}{k}, \frac{-1}{k}\right] \\ &= \mathcal{M}_{+t} \circ [1, t, -k] = [1, t-2t, -k-t^2+t^2] = [1, -t, -k] \end{aligned}$$

Where again we have used the ‘dividing out by the leading coefficient’ technique.

As trivial of a procedure as this may seem, I find it very fascinating to think of the roots of these polynomials being in a kind of infinite *superposition*; trapped by the symmetries of the coefficients into being equal to a specific transformation upon themselves.

If a polynomial has roots r which obey the symmetry $r = \mathcal{M}_{f_1 \circ f_2 \dots f_n} \circ r$, then clearly that polynomial also has roots $\mathcal{M}_{f_1 \circ f_2 \dots f_n} \circ \mathcal{M}_{f_1 \circ f_2 \dots f_n} \circ r$, and so on like an infinite set of stacking dolls. Since a polynomial can only have a finite number of roots, we conclude that these roots are candidate values for the recursive object defined by the same $f_1 \circ f_2 \dots f_n$ invariance.

Ultimately, it is up to the conditions of fixed-point iteration, not discussed here, whether any such root actually has a basin about which a given initialized value x_0 converges under $x_{i+1} = f_1 \circ f_2 \dots f_n \circ x_i$.

Resultants & Quartics

By now, the notion actions on polynomial roots x, y as representations for actions on recursive objects X, Y should be a familiar one. We now turn to the process of representing $X + Y, XY$, and other similar algebraic combinations as new recursive objects by finding minimal, resultant polynomials for the quantities $x + y, xy$, etc.

For example, supposing we take two linear polynomial representatives $X = a + b(a + b(\dots)) \Rightarrow X = a + bX \Rightarrow X = \frac{a}{1-b}$ and $Y = c + d(c + d(\dots)) \Rightarrow Y = c + dY \Rightarrow Y = \frac{c}{1-d}$, we can reformulate their sum in terms of a new recursive object;

$$\begin{aligned} X + Y &= \frac{a}{1-b} + \frac{c}{1-d} = \frac{a(1-d) + c(1-b)}{(1-b)(1-d)} = \frac{a+c-ad-bc}{1-(d+b-bd)} \\ &= [a+c-ad-bc] + [d+b-bd]([a+c-ad-bc] + [d+b-bd](\dots)) \end{aligned}$$

In the same fashion, their product also can be expressed in the recursive form;

$$XY = \frac{ac}{(1-b)(1-d)} = \frac{ac}{1-(d+b-bd)} = [ac] + [d+b-bd]([ac] + [d+b-bd](\dots))$$

Results not so easily seen without the aid of a polynomial representation. As a purely quadratic example, consider the following constants and their continued-fraction representations;

$$\begin{aligned} x &= 1 + \sqrt{3} \Rightarrow x^2 - 2x - 2 = 0 \Rightarrow X = 2 + \frac{2}{2 + \frac{2}{\dots}} \\ y &= 2 + \sqrt{3} \Rightarrow y^2 - 4y + 1 = 0 \Rightarrow Y = 4 - \frac{1}{4 - \frac{1}{\dots}} \end{aligned}$$

The minimal polynomial for $z = x + y$ is readily found and can then be used to represent $Z = X + Y$;

$$z = x + y = 3 + 2\sqrt{3} \Rightarrow z^2 - 6z - 3 = 0 \Rightarrow Z = 6 + \frac{3}{6 + \frac{3}{\dots}}$$

And an identical process for $w = xy, W = XY$ yields;

$$w = xy = 5 + 3\sqrt{3} \Rightarrow w^2 - 10w - 2 = 0 \Rightarrow W = 10 + \frac{2}{10 + \frac{2}{\dots}}$$

Thus, without much effort at all we have established a *field* over all integer continued fractions which share a common irrational, algebraic component! ($\sqrt{3}$ in the above example.) It is worth noting that the fraction $1 = 1 + \frac{0}{1 + \frac{0}{\dots}}$ is a multiplicative identity element and $0 = \lim_{\epsilon \rightarrow 0} \epsilon + \frac{0}{\epsilon + \frac{0}{\dots}}$ is a kind of additive identity element.

Indeed, if it is stipulated that associated polynomials must have integer coefficients, the sum of two

quadratically-defined roots will, in general, require a quartic representation.

To aid in this exploration, the following recursively-defined object is equivalent to the *fully* general quartic $x^4 + ax^3 + bx^2 + cx + d = 0$;

$$X = -p + \sqrt[3]{p \left(-\frac{a^2}{2} + \frac{4}{3}b \right) + \frac{1}{6}ba - c + \frac{p \left(-\frac{3}{8}a^3 + \frac{3}{2}ba - 3c \right) + \left(\frac{1}{8}ba^2 - \frac{3}{4}b^2 + ac - d \right)}{a - 3p + X}}$$

Where $p = \frac{3a + \sqrt{9a^2 - 24b}}{12}$.

Combining together the addition and multiplication schemes presented here and the functional operations from previous sections can give all kinds of interesting results, for example we have;

$$\left(1 + \frac{6}{1 + \frac{6}{1 + \frac{6}{\dots}}} \right)^2 + \left(2 + \frac{8}{2 + \frac{8}{2 + \frac{8}{\dots}}} \right)^2 = \left(4 + \frac{5}{4 + \frac{5}{4 + \frac{5}{\dots}}} \right)^2$$

The reader is invited to ponder which *constraints* on the entries in a given recursive object permit it's roots to lie within the space of say $p + qs, p, q \in \mathbb{Q}, s \notin \mathbb{Q}$, at which point addition and multiplication of the objects is assured to be *closed*.

Again, rather than exhausting every possible avenue by which one could combine together algebraic combinations of recursive objects, I hope it is clear from these simple linear and quadratic examples how further conclusions might be drawn via the process of finding polynomial resultants.

Linear Transformations

Yet another lens through which continued fractions can be seen as very *natural* recursive objects is by considering how their generating transformations, $1/$, $\times k$, and $+t$, are *linear* in the coefficients of $p(x) = Ax^2 + Bx + C$. Specifically, taking one recursive ‘step’ on the roots of p , ie $r_{i+1} = t + k/r_i$, is simply;

$$\mathcal{M}_{+t} \circ \mathcal{M}_{\times k} \circ \mathcal{M}_{1/} \circ [A, B, C] = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ -2t & 1 & 0 \\ t^2 & -t & 1 \end{bmatrix}}_{\text{Shifting}} \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & k & 0 \\ 0 & 0 & k^2 \end{bmatrix}}_{\text{Scaling}} \underbrace{\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}}_{\text{Reversal}} \begin{bmatrix} A \\ B \\ C \end{bmatrix} = \mathbf{M}_{t,k} \begin{bmatrix} A \\ B \\ C \end{bmatrix}$$

It should be perfectly plausible that the vector $[-1/k, t/k, 1]$ is an *eigenvector* of $\mathbf{M}_{t,k}$; for we have already seen that the coefficients of the associated polynomial to the transformation $\mathbf{M}_{t,k}$, ie $[-1/k, t/k, 1] \sim [1, -t, -k]$ must be stationary under its action.

Given an *infinite* sequence of $(t_1, k_1), (t_2, k_2) \dots$, it was my intention to extend our study of *recursive* continued fractions to non-periodic ones.

Indeed, should there exist a sequence of vectors $v_n, v_{n-1}, \dots, v_1$ such that $\mathbf{M}_{t_{i+1}, k_{i+1}} v_{i+1} = v_i$, then;

$$\mathbf{M}_{t_1, k_1} \mathbf{M}_{t_2, k_2} \dots \mathbf{M}_{t_{n-1}, k_{n-1}} \mathbf{M}_{t_n, k_n} v_n = \mathbf{M}_{t_1, k_1} \mathbf{M}_{t_2, k_2} \dots \mathbf{M}_{t_{n-1}, k_{n-1}} v_{n-1} = v_1$$

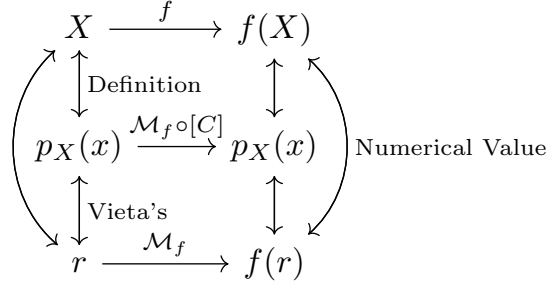
And so the roots of the polynomial corresponding to v_1 (which may or may not have rational coefficients) would provide a value for the fraction $t_1 + \frac{k_1}{t_2 + \frac{k_2}{t_3 + \frac{k_3}{\dots}}}$, assuming $\lim_{n \rightarrow \infty} v_n / |v_n|$ exists.

Finding the v sequence seems to be equivalent in difficulty to finding a recursion on the numerators and denominators at each step of the fraction, i.e. $x = t_1 + \frac{k_1}{x}$, $x = t_1 + \frac{k_1}{t_2 + \frac{k_2}{x}}$, ..., giving no ‘new’ benefit to the approach of matrices.

In addition, even by scaling each $\mathbf{M}_{t_i, k_i}$ by a normalization constant, the convergence of the *roots* of the resulting polynomial do not seem to guarantee a convergence, or even finitude, of the polynomial *coefficients*, as was hoped.

Conclusion Diagram & Exercises

Calling X our recursive object, p_X its associated polynomial, C the coefficients of that polynomial, and r the set of its roots, we can summarize the discussions of this note with the following diagram:



In addition to the many intriguing identities formed from manipulating recursive objects, it is this *interconnectedness* between the root representations, coefficient representations, and polynomial representations of these entities which give this topic interest.

Finally, we conclude this note with the following set of helpful exercises of which, I would *especially* recommend the **Dual Generators** and **Characteristic Convergents** problems: these last two present additional ideas not discussed in the above sections that are particularly motivated to discover on ones own.

Exercises:

1. (**Doubly Continued Fraction**) Find and solve the quadratic equation associated with;

$$X = \frac{2 + \frac{2 + \frac{2 + \dots}{3 + \dots}}{3 + \frac{2 + \dots}{3 + \dots}}}{3 + \frac{2 + \frac{2 + \dots}{3 + \dots}}{3 + \frac{2 + \dots}{3 + \dots}}}$$

Thus determining its exact numerical value.

2. (**Cubic Recursion**) Verify the following party trick by finding an associated *cubic* polynomial.

$$\cos(x) = \sqrt{\frac{3}{4} + \frac{\frac{\cos(3x)}{4}}{\sqrt{\frac{3}{4} + \frac{\frac{\cos(3x)}{4}}{\sqrt{\frac{3}{4} + \frac{\cos(3x)}{4}}}}}}$$

3. (**Very Weak Fixed Point Iteration Theorem**) Suppose f has the property that $\forall x, y \in I \subseteq \mathbb{R}, |f(x) - f(y)| \leq C|x - y|$ where $C < 1$ and that I contains exactly one solution to $a = f(a)$. Prove that the iteration $x_{i+1} = f(x_i)$ leads to $\lim_{i \rightarrow \infty} x_i = a$.
4. (**Root Squaring Mapping**) Verify the coefficient mapping $\mathcal{M}_{\square^2}$ by comparing coefficients between $x^2 + A'x + B' = (x - r_1^2)(x - r_2^2)$ and $x^2 + Ax + B = (x - r_1)(x - r_2)$. Hint: Use $(r_1 + r_2)^2$.

In addition, confirm that for $r_1 \geq 0, r_2 \geq 0$, composing $\mathcal{M}_{\sqrt{\cdot}} \circ \mathcal{M}_{\square^2} = \mathcal{M}_{\square^2} \circ \mathcal{M}_{\sqrt{\cdot}} = \mathcal{I}$ gives the identity transformation.

5. (**Quartic Field**) Convince yourself informally why adding a ‘new square root’, again an informal term, requires a quartic minimal polynomial. For example, if we wanted to find a polynomial with root of $2\sqrt{3} + 3\sqrt{7}$, we would consider;

$$(x - 2\sqrt{3} - 3\sqrt{7})(x - 2\sqrt{3} + 3\sqrt{7})(x + 2\sqrt{3} - 3\sqrt{7})(x + 2\sqrt{3} + 3\sqrt{7})$$

Since the resulting polynomial must have rational coefficients, *each* irrational root must be present with its *conjugate*.

6. (**Polynomial Shift Operator**) Consider the ‘shifting’ matrix discussed in the *Linear Transformations* section. Generalize the $\mathcal{M}_{+t}$ matrix to polynomials of arbitrary degree. Where do the coefficients on the columns come from?

Clearly it is true that $\mathcal{M}_{+t}^{-1} = \mathcal{M}_{-t}$. However, looking only at the coefficients in the matrix, how might you prove this?

7. (**Dual Generators**) If x is a fixed point of $f_1 \circ f_2 \circ \dots \circ f_n$, show that x is also a fixed point of $f_n^{-1} \circ \dots \circ f_2^{-1} \circ f_1^{-1}$.

Thus, if $\mathcal{M}_{+t} \circ \mathcal{M}_{\times k} \circ \mathcal{M}_{1/}$ is the ‘standard’ set of generators for a continued fraction, X , show that the following ‘dual continued fraction’, $\overline{X}$, generated by the inverse map;

$$\overline{X} = \frac{k}{\frac{k}{\dots - t} - t}, \quad X = t + \frac{k}{t + \frac{k}{t + \dots}}$$

Is *also* a root of $x^2 - tx - k$. In fact the dual $\overline{X}$ object takes the value of the (often negative) root otherwise *rejected* when finding the value for X deemed to be extraneous.

Ignoring concerns about convergence, find the dual object to;

$$X = \sqrt{b + a\sqrt{b + a\sqrt{b + a\sqrt{\dots}}}}$$

By studying the inverse of $\mathcal{M}_{\sqrt{}} \circ \mathcal{M}_{+b} \circ \mathcal{M}_{\times a}$.

8. (**Characteristic Convergents**) The *convergents*, c_n , of a continued fraction are the rational approximations formed from the first n divisions. For example, the convergents for $X = 1 + \frac{1}{1 + \frac{1}{1 + \dots}}$ are $c_1 = 1 + \frac{1}{1}$, $c_2 = 1 + \frac{1}{1 + \frac{1}{1}}$, $c_3 = 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1}}}$, etc.

Using induction, show that for $n \geq 3$, the n th convergent of X is a ratio of consecutive Fibonacci numbers.

Since the limit of the convergents as $n \rightarrow \infty$, assuming it exists, is *defined* to be the numerical value for the continued fraction, conclude that;

$$\lim_{n \rightarrow \infty} \frac{F_n}{F_{n-1}} = \varphi, \varphi^2 - \varphi - 1 = 0$$

That φ satisfies the equation $x^2 - x - 1$ is especially suspicious given that the *characteristic* polynomial for the Fibonacci sequence is;

$$F_{n+2} = F_{n+1} + F_n \rightarrow r^{n+2} = r^{n+1} + r^n \Leftrightarrow r^2 - r - 1 = 0$$

Where the characteristic polynomial is a standard tool ‘formed’ from ‘assuming’ a geometric progression for F_n .

Indeed, any sequence defined by a linear two-step recursion can be *encapsulated* as a continued fraction a similar inductive argument;

$$y_{n+2} = \alpha y_{n+1} + \beta y_n \Leftrightarrow \alpha + \frac{\beta}{\frac{y_{n+1}}{y_n}} = \frac{y_{n+2}}{y_{n+1}} \Leftrightarrow Y = \alpha + \frac{\beta}{\alpha + \frac{\beta}{\alpha + \dots}}$$

Prove that the *characteristic* polynomial for such a sequence matches the *associated* polynomial for the resulting continued fraction Y .